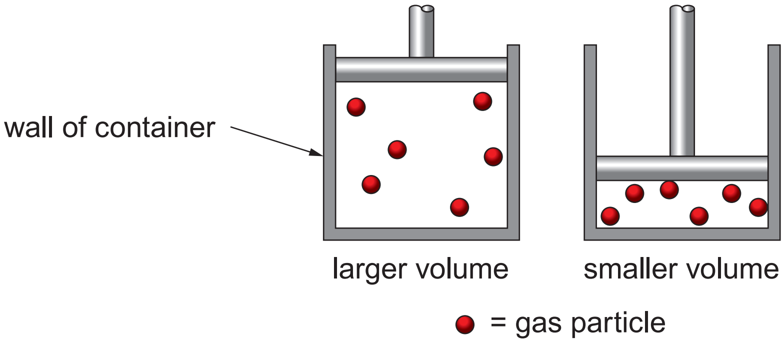
Examiner only

6. Kinetic theory is used to explain the behaviour of gases.

(a) A fixed mass of gas is kept at a constant temperature in a container.
Explain why the pressure increases when its volume is decreased. [3]



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(b) Students are given data about the behaviour of a fixed mass of gas kept at **constant pressure**. The volume of the gas changes with temperature as shown in the table below.

Temperature (°C)	Volume (cm^3)
−223	20
−173	40
−73	80

Examiner only

(i) Chris states that

$$\frac{\text{volume}}{\text{temperature}} = \text{constant}$$

when temperatures are measured in °C. Explain whether the results agree. [2]
Space for calculations.

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(ii) The students plot a graph of the results opposite. Explain how they can use the graph to determine the value of absolute zero in °C. [2]

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(c) A balloon is filled to a volume of 2 800 cm^3 at 7 °C (280 K).
The balloon is heated to a temperature of 67 °C.
The pressure remains constant.

Use the equation:

$$\frac{pV}{T} = \text{constant}$$

to calculate the new volume of the balloon. [4]
[T (K) = θ (°C) + 273]

volume = ... cm^3

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